Peter Dietrich
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September 15, 2023
NRC-23-0052

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2023-002-00

Pursuant to Title 10 Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), DTE Electric Company (DTE) is submitting LER No. 2023-002-00 "Technical Specification Completion Time Exceeded Due to Repairs on the MDCT Fan D".

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2023-002-00, "Technical Specification Completion Time Exceeded Due to Repairs on the MDCT Fan D"

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

Enclosure to
NRC-23-0052

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2023-002-00
“Technical Specification Completion Time Exceeded Due to Repairs on the MDCT Fan D”

NRC FORM 366 (09-01-2023)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 09/30/2023

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
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4. Title
Technical Specification Completion Time Exceeded Due to Repairs on the MDCT Fan D.

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
07	21	2023	2023	- 002 -	00	09	19	2023	NA	☐ 050
									NA	☐ 052

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Eric Frank - Manager, Nuclear Licensing	734-586-4772

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	CC	SPT	X999	Y	B	CC	VB	M085	Y

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 0424 Eastern Daylight Time (EDT) on July 18, 2023, the Division II Mechanical Draft Cooling Tower (MDCT) fan "D" was declared inoperable due to a trip of the fan motor while running at high speed. The MDCT fans are required to support operability of the Ultimate Heat Sink (UHS). The UHS is required to support operability of the Division II Emergency Equipment Cooling Water (EECW) system and the Emergency Equipment Service Water (EESW) system. The cause of the event was either the degraded mounting hardware between the gear reducer and the pedestal, and/or the degraded bushings on the driveshaft coupling to the gear reducer. The necessary repairs extended beyond the end of the 72-hour completion time prescribed by Technical Specification (TS) 3.7.2. A Notice of Enforcement Discretion (NOED) was approved by the NRC on July 20th, 2023, expiring at 2324 on July 25, 2023. This event is being reported under 10 CFR 50 .73(a)(2)(i)(B) for operation or condition prohibited by TS. Repairs were completed on the degraded mounting hardware and the degraded bushings, and the Division II MDCT fan "D" was declared operable at 1557 on July 22, 2023.

There were no radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

INITIAL PLANT CONDITIONS
Mode – 1
Reactor Power – 100

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 0424 Eastern Daylight Time (EDT) on July 18, 2023, the Division II Residual Heat Removal Service Water (RHRSW) [BI] Mechanical Draft Cooling Tower (MDCT)[CTW] Fan “D” [FAN], was declared inoperable due to a trip of the fan motor while running at high speed.

The MDCT fans are required to support operability of the Ultimate Heat Sink (UHS) [BS]. The UHS reservoir is divided into two, one-half capacity reservoirs, corresponding to Division I and Division II. A two-cell MDCT is located above each of the one-half capacity reservoirs. Each cell is equipped with a MDCT fan. Two MDCT fans above each one-half capacity reservoir are required for it to be considered operable. Each reservoir is the cooling source for that division's service water subsystem (e.g., the Emergency Diesel Generators (EDGs) [DG] cooling water [LG], Emergency Equipment Cooling Water (EECW) system [CC], Emergency Equipment Service Water (EESW), RHR [BO] system, Core Spray [BG] system). The Division II EECW system cools the High-Pressure Coolant Injection (HPCI) [BJ] system room cooler [CLR]. Typically, when Division II UHS is declared inoperable, then the high-pressure coolant injection (HPCI) system is declared inoperable; however, an Engineering analysis has been performed for HPCI operability which determined that there is reasonable assurance that HPCI would have performed its design function for its given mission time and remained operable.

Investigation found that the cause of the trip was due to high vibrations caused by either the degraded mounting bolt (one of three) of the gear reducer to the pedestal [SPT] and/or the degraded bushings [VB] (Manufacturer Marley Company, Part # 69-29-02.) on the driveshaft coupling to the gear reducer. Further analysis is being done to determine the cause of the degradation on those parts. Based on the equipment history of the MDCT fans and the cause of the failure, there is no firm evidence to identify that the failure would have existed previously. Based on NUREG-1022 Section 3.2.2 guidance, the failure is assumed to have occurred at the time of discovery.

The necessary repairs extended beyond the end of the 72-hour completion time prescribed by REQUIRED ACTION A.1 of Technical Specification (TS) 3.7.2.(Emergency Equipment Cooling Water (EECW), Emergency Equipment Service Water (EESW) and Ultimate Heat Sink (UHS) – one reservoir inoperable). A Notice of Enforcement Discretion (NOED) was approved by the NRC on July 20, 2023, effective from 0424 EDT on July 21, 2023, to 2324 EDT on July 25, 2023. The enforcement discretion provided authorization for temporary non-compliance with TS 3.7.2 CONDITION A as described above. In addition, the following TS were also exceeded, TS 3.8.7 CONDITION A (Distribution Systems Operating, one or more required AC Electrical Distribution subsystems inoperable), and CONDITION B (one or more required DC Electrical Distribution subsystems inoperable), TS 3.8.4 CONDITION B (DC sources operating, one DC Electrical Power Subsystem inoperable) and TS 3.8.1 CONDITION B (AC Sources Operating, both EDGs in one division inoperable).

Following the degraded bushing replacement and removal and regrouting of the gearbox foundation bolts, the MDCT Fan “D” was declared operable at 1557 EDT on July 22, 2023.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER: YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

Per NRC Enforcement Policy Appendix “F”, NRC approval of discretionary enforcement does not relieve the licensee from reporting requirements. This event is being reported under 10 Code of Federal Regulation (CFR) 50 .73(a)(2)(i)(B) for operation or condition prohibited by TS.

Since the Division I MDCT fans were operable, Division I of UHS and all its supported equipment were operable throughout this event. As a result, there was no loss of safety function associated with the UHS itself. The mission time associated with HPCI is less than 24 hours. An engineering analysis has shown that use of a single MDCT fan in one division can achieve acceptable results for a 24-hour period for that division. The conditions present during this event, where Division II UHS had one operable MDCT fan (Fan "B") were less challenging than the conditions utilized in the engineering analysis. Therefore, the short period of time where MDCT Fan "D" was inoperable, would have had little or no impact on the functionality of HPCI.

There were no safety consequences associated with this event. There were no radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

CAUSE OF THE EVENT

The cause of exceeding TS 3.7.2 Condition A completion time was due to the amount of time necessary to make the repairs to the MDCT Fan "D" and restore operability.

The direct cause of the Division II MDCT Fan “D” tripping on high vibrations was either the degraded mounting condition of the gear reducer to its pedestal or the degraded bushings on the driveshaft coupling to the gear reducer (or both). Either the front bolt on the gear reducer mount failed first or the bushings on the driveshaft coupling to the gearbox failed first due to their degraded condition. In either case, the failed component allowed motion to be introduced to the system while running, which is believed to have caused the other component to fail. The failure of the gear reducer mounting bolt caused the gear reducer to not be rigid to the foundation. This allowed the gear reducer to have vertical movement while operating, which introduced vibration to the fan, which caused the unit to trip offline.

CORRECTIVE ACTIONS

The timeline to make the necessary repairs described below exceeded TS 3.7.2 Condition A completion time, therefore a NOED was completed, and approval was granted by the NRC.

At 1556 on July 22, 2023, repairs were completed to the pedestal mounting hardware and the degraded bushings. Post Maintenance Testing (PMT) was completed, and the Division II MDCT fan “D” was placed back in service. During the “extent of condition” review, it was discovered that MDCT fan “A” and “C” pedestals were also degraded and non-conforming, but remained operable, and in need of similar repair. To be proactive, an exigent LAR was submitted to the NRC on August 10th, 2023 (NRC-23-0050), requesting a one-time extension of TS 3.7.2 to repair Division I MDCT Fan “A” and “C” online by November of 2023.

NARRATIVE

PREVIOUS OCCURRENCES

There have been no previous similar occurrences.